CORAL REEF HEALTH PREDICTION USING DEEP LEARNING

A PROJECT REPORT

(22AIE304 Deep Learning)

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Under the guidance of

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ABSTRACT

Reef surveys produce thousands of quadrat photographs, and in most archives the only labels are sparse points that an expert has clicked and named. We use those points directly. From a public collection of 2,455 annotated quadrat photographs carrying 404,255 valid expert points, we group 40 raw benthic labels into three classes (Healthy Coral, Dead/Degraded and Substrate/CCA) and train every model on 224 × 224 crops centred on the labelled points. Six models are compared under one protocol: four ImageNet-pretrained CNNs (MobileNetV2, ResNet18, EfficientNet-B0, ShuffleNetV2), a residual squeeze-and-excitation CNN trained from scratch (CoralCNN-SE), and a DINOv2 ViT-S/14 with its last four blocks fine-tuned. The split is by photograph, so no reef image appears in more than one set.

On 6,000 held-out points DINOv2 is the most accurate model (86.70% accuracy, macro-F1 0.831, ROC-AUC 0.956). ResNet18 follows at 85.97% and 0.819, and the scratch CNN reaches 83.68% with the highest Dead/Degraded recall (0.745). Sliding each classifier over 150 held-out quadrats and blending the overlapping windows gives a dense health map. The DINOv2 map estimates live-coral cover with a mean absolute error of 3.11 percentage points ($r = 0.981$) and puts 92% of quadrats in the same reef-status band as the expert count. Stronger augmentation, CutMix/MixUp and focal loss all lowered Dead/Degraded recall compared with a plain baseline. Grad-CAM finds the labelled point for ResNet18 (peak within 32 px in 91% of crops) but not for DINOv2 (11%), and occlusion sensitivity was the most faithful explanation method for all three model families. Dead/Degraded remains the weak class, with F1 between 0.67 and 0.71.

Keywords: coral reef monitoring, benthic classification, transfer learning, squeeze-and-excitation, DINOv2, sliding window, Grad-CAM, explainability

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LIST OF SYMBOLS AND ABBREVIATIONS

AMP	-	Automatic mixed precision
AP	-	Average precision (area under the precision-recall curve)
AUC	-	Area under the curve
BN	-	Batch normalisation
CAM	-	Class activation map
CCA	-	Crustose coralline algae
CLAHE	-	Contrast-limited adaptive histogram equalisation
CNN	-	Convolutional neural network
ECE	-	Expected calibration error
GFLOPs	-	Giga floating-point operations per forward pass
Grad-CAM	-	Gradient-weighted class activation mapping
IG	-	Integrated Gradients
MAE	-	Mean absolute error
MCC	-	Matthews correlation coefficient
pp	-	Percentage points
RMSE	-	Root mean squared error
ROC	-	Receiver operating characteristic
SE	-	Squeeze-and-excitation
TL	-	Transfer learning
TTA	-	Test-time augmentation
ViT	-	Vision transformer
XAI	-	Explainable artificial intelligence
κ	-	Cohen's kappa
π_c	-	Prior probability of class c
τ	-	Strength of the prior-shift correction
r	-	Pearson correlation coefficient

CHAPTER 1
INTRODUCTION

1.1 BACKGROUND

Coral reefs are monitored mainly through one number: the share of the sea floor covered by live coral, tracked at fixed sites year after year. The standard way to measure it is the photo-quadrat. A diver photographs a framed square of reef, and later an analyst overlays points on the photograph and names what lies under each one, whether a coral genus, turf algae, sand or rubble. Percent cover is the share of points in each category.

The photography is quick. The annotation is not. The photographs used in this project carry about 165 points each, and every point is a separate decision by a trained person. Survey programmes therefore hold far more photographs than they can label, and the labels they do have are sparse points, not outlines or masks.

Deep learning can take over the naming step, and much recent work tries to. A large part of it treats reef health as whole-image classification, healthy against bleached, on a few hundred photographs collected from the web (Macrohon and Hung, 2025; Navaz, 2026). A survey quadrat does not fit that setup. It is a mixed scene with coral, algae and bare substrate in one frame, and the question a reef manager asks is how much of each is present, not which single label the photograph deserves.

1.2 MOTIVATION

Our own earlier attempt, presented at Review 2, ran into this problem. We gave each quadrat the label of its most common point class and trained on whole photographs. Pretrained CNNs reached 81 to 84% accuracy, a CNN trained from scratch stalled at 65.7%, and macro-F1 stayed between 0.52 and 0.76 because the minority classes were mostly lost.

The annotations explain why. Of the 2,455 photographs, 88.1% contain some live coral, yet only 17.1% would be labelled Healthy Coral by majority vote, and 68.1% contain all three classes. A whole-image label throws most of that information away and teaches the network that a photograph with a quarter of its area in coral is “substrate”. That observation shaped the final design: classify what is under a point, then build the image-level answer from many such decisions.

1.3 OBJECTIVES

1. Build a point-level data set for three-class reef health classification from sparse expert annotations, with no photograph shared between training and test sets.

2. Compare transfer-learning CNNs, a CNN trained from scratch and a vision transformer on identical crops, split and metrics.

3. Turn the point classifier into a dense reef map and a live-coral cover estimate with a sliding window, and measure how close the estimate is to the expert count.

4. Handle the difference in class frequencies between the balanced training set and real reef data, and test whether heavier augmentation or loss re-weighting helps the minority class.

5. Report cost next to accuracy: parameters, FLOPs, GPU and CPU latency, and time per quadrat.

6. Check model explanations quantitatively against the position of the expert point, using more than one explanation method.

1.4 CONTRIBUTIONS

The report makes four contributions. The first is a like-for-like benchmark of six models on 24,000 class-balanced training crops and 6,000 naturally distributed test points. In it the from-scratch CNN closes most of the gap to the pretrained models (83.7% against 86.0% for ResNet18). The second is a sliding-window evaluation on 150 whole quadrats that scores the dense map at 25,082 expert points and as percent cover; the best model is within 3.11 percentage points of the expert cover on average.

The third is a negative result. Class-aware strong augmentation, CutMix/MixUp and focal loss all did worse than the plain training recipe, while a prior correction tuned on the validation set gave a controllable trade-off between overall accuracy and recall of the Dead/Degraded class. The fourth is a quantitative comparison of Grad-CAM, Integrated Gradients and occlusion sensitivity on three model families, with the labelled point as the location reference.

1.5 ORGANISATION OF THE REPORT

Chapter 2 reviews the literature and states the gaps this project works on. Chapter 3 covers the problem, the data and the evaluation protocols, Chapter 4 the models and methods, Chapter 5 the results, and Chapter 6 the conclusions and further work.

CHAPTER 2
LITERATURE REVIEW

We reviewed 26 papers: six foundational ones and twenty from December 2024 to 2026. They are grouped below by the part of our pipeline they inform. Each section closes with what the papers leave open.

2.1 FOUNDATIONAL WORK

He et al. (2016) showed that identity skip connections let a layer learn a residual function, so networks of up to 152 layers train without the accuracy degradation seen in plain deep stacks. Their ensemble reached 3.57% top-5 error on ILSVRC 2015. Hu et al. (2018) added the squeeze-and-excitation (SE) block, which pools each channel to a single value and passes the result through two small layers and a sigmoid to re-weight the channels. SE networks won ILSVRC 2017 at 2.251% top-5 error, and SE-ResNet-50 cut top-5 error from 7.48% to 6.62% for 0.26% more computation. Both ideas are in our CoralCNN-SE, and ResNet18 is one of our transfer backbones.

Ma et al. (2018) designed ShuffleNet V2 from speed measured on the device instead of FLOPs, and argued that the two do not rank networks the same way. We report both and see the same mismatch. Yosinski et al. (2014) studied how well features transfer layer by layer: transferred features help even after fine-tuning, but less as the source and target tasks move apart. ImageNet objects and reef texture are far apart, and the size of the benefit for point-level reef labels had not been measured.

Selvaraju et al. (2017) introduced Grad-CAM, which weights the last convolutional feature maps by the gradient of a class score to produce a heat-map without retraining. It was validated on ImageNet localisation (56.51% top-1 localisation error with VGG-16) and on human studies with object photographs. Oquab et al. (2024) trained DINOv2, a self-supervised vision transformer, on 142 million curated images and showed that its frozen features work with simple heads across classification, segmentation and depth benchmarks. Underwater benthic imagery was not among them.

2.2 CORAL REEF HEALTH AND BENTHIC CLASSIFICATION

Three recent papers classify coral as healthy or bleached from whole images. Macrohon and Hung (2025) compared a CNN, ResNet-50 and a ViT on 923 Flickr images and reported

88%, 86% and 64% accuracy. Navaz (2026) used the same 923 images with ConvNeXt-Tiny and Swin-Tiny (both 86.33%), ResNet50 (84.17%) and EfficientNetB3 (83.45%). Menaga et al. (2025) fine-tuned MobileNetV2, VGG16 and ResNet and found ResNet the most accurate and MobileNetV2 the best balance of accuracy and time, though the accessible abstract gives no figures. All three are binary, use fewer than a thousand images, and produce one label per photograph with no spatial output.

Larger efforts exist but aim at different targets. Shao et al. (2025) adapted DINOv2 with LoRA to 1,203 photographs from 15 dive sites at Koh Tao, cut into 42,105 patches with eight health and stressor labels. They reached a 64.77% match ratio and 88.05% micro-F1 with 5.91 M trainable weights, against 60.34% for the best conventional model, and noted a large accuracy drop between wet and dry seasons. ReefNet (Felemban et al., 2025) collects about 925,000 genus-level point annotations over 181,223 images from 76 CoralNet sources. A fine-tuned ViT-B/16 gets 79.84% macro recall within a source, and the best cross-source result is 59.80%. Coralscapes (Sauder et al., 2025) provides dense masks for 2,075 Red Sea images in 39 classes, and its best model, SegFormer, reaches 57.80 mIoU.

None of these gives a three-class health reading from ordinary survey points. ReefNet targets coral genus, Coralscapes needs dense masks that most archives do not have, and Shao et al. report patch labels without a cover map.

2.3 POINT ANNOTATIONS, PATCHES AND SLIDING WINDOWS

Blondin et al. (2024) built a hierarchical classifier over the benthic label tree, with an EfficientNet-B0 pretrained on 16 million benthic images and MLP heads, on 1,549 images of 25 points each. The gain over a flat classifier was about 2% F1. Mohamed and Nadaoka (2026) placed 56-pixel patches at superpixel centres, upscaled them, and fed ConvNeXt, EfficientNet and DINOv2 features to an SVM. They reported 96 to 97% overall accuracy on four orthomosaic sets, but with only 500 to 600 samples per site.

Borja et al. (2026) tried to turn sparse points into dense masks with SAM2 and then trained SegFormer on the result. Scores rose on their own benchmark (27.53 to 32.00) and on Coralscapes, but not on UCSD Mosaics, and they reported masks that bleed across class boundaries. Trotter et al. (2025) sliced Antarctic sea-floor images into 500-pixel patches with 50% overlap for object detection. The best detector reached mAP@0.5 of 0.49 for medium and large organisms, and slicing struggled with objects that span several patches.

Two lessons carry over. A label is only exact at the point where the expert put it, so we centre each training crop on that point and do not expand it into a region. And a patch classifier needs overlap and blending when it is slid across an image, or the seams show.

2.4 TRANSFER LEARNING AND LIGHTWEIGHT CNNS

Shahriar (2025) fine-tuned seven lightweight CNNs under one protocol on CIFAR-10, CIFAR-100 and Tiny ImageNet. EfficientNetV2-S was best on all three, and EfficientNet-B0 came close on Tiny ImageNet with about a fifth of the parameters. Akhand (2026) compared a custom VGG-style CNN with frozen and fine-tuned VGG-16 on five image sets of 450 to 14,000 images. Fine-tuning won everywhere (92.1 to 96.8% against 89.8 to 94.2% for the custom network) at about twice the time per epoch. AQUA20 (Fuad et al., 2025) benchmarks 13 models on 8,171 underwater images in 20 marine classes, with ConvNeXt at 90.69% top-1, and shows Grad-CAM and LIME maps for visual inspection.

These comparisons use balanced, generic benchmarks or species labels. None reports macro-F1 under class imbalance, and the custom networks they test have neither residual connections nor attention.

2.5 RESIDUAL NETWORKS WITH SQUEEZE-AND-EXCITATION

Evidence for SE in residual networks comes mostly from outside ecology. Farisi (2025) added SE to every block of ResNet-18, pruned it, and gained 0.9 points on CIFAR-10 (94.1% against 93.2%) with 27% fewer parameters. Banyal and Kumar (2026) fine-tuned MobileNetV2-SE and ResNet50-SE on four mammography data sets; MobileNetV2-SE scored 92.35 to 99.14%, while ResNet50-SE ranged from 71.60% to 100% depending on the data set. He et al. (2025) combined an inverted residual network, an efficient SE module and two-step transfer learning for breast tumour classification and reported a mean accuracy of 0.818.

The one reef study is Harrison et al. (2025), who used a ResNet101 U-Net with spatial and channel SE on 125,901 tiles from 77 orthomosaics in Kiritimati. They reached 84% overall accuracy and an mIoU above 0.8, against 0.48 for plain U-Net, but the method needs 6 to 10 hours of dense annotation per plot and confuses the different forms of dead coral. Whether a small residual SE network can be trained from scratch on reef photographs with only point labels was open.

2.6 EXPLAINABILITY

Panboonyuen (2026) scored Grad-CAM on five architectures for lung CT classification using localisation accuracy and perturbation tests. The maps highlighted tumours for most CNNs but degraded for the ViT-B/16. Bent et al. (2026) wrote guidance on explainable AI for ecological images with two case studies, seal detection and whale segmentation, in which explanations exposed reliance on dark ice, water texture and image-edge artefacts. They stress that a plausible map is not necessarily a faithful one and that ecology lacks benchmarks for judging explanations. Ibrahim and Muruganantham (2025) proposed a two-stage ViT for white band disease with Grad-CAM, but the accessible page gives neither a data set nor metrics.

In reef work, explanations have been shown as pictures and not scored. Point annotations make a score possible, because each crop comes with the exact pixel the expert labelled.

2.7 RESEARCH GAPS

Read together, the 26 papers leave six gaps that this project takes up.

1. Reef-health classifiers are mostly binary and trained on about 900 whole images. We classify three classes at point level from 404,255 expert points.

2. Dense masks are costly, and expanding points into masks is noisy. We train and evaluate from sparse points alone and still produce a map and a cover estimate.

3. CNNs and transformers have not been compared at patch level on identical reef labels. Our six models share crops, split, training recipe and metrics.

4. SE and residual designs have been tested on CIFAR-10, mammography and dense reef segmentation, but not from scratch on small point-labelled reef data.

5. Explanations for reef models have not been scored against expert labels. We score three methods on three model families.

6. Cost is seldom reported next to accuracy. We measure parameters, FLOPs, GPU and CPU latency and time per quadrat for every model.

Two gaps raised by this literature stay open in our work. Shao et al. (2025) and Felemban et al. (2025) both report large drops across sites and seasons, and we do not test on a second, independent reef. We also have no SE on/off ablation, so we cannot say how much the SE gate itself contributes.

CHAPTER 3
PROBLEM STATEMENT AND METHODOLOGY

3.1 PROBLEM STATEMENT

Given a quadrat photograph and nothing else, the system must do two things. It must predict which of three benthic classes (Healthy Coral, Dead/Degraded or Substrate/CCA) lies under any pixel of the photograph. And it must estimate, from those predictions, the percentage of the quadrat covered by each class, live coral above all.

The only supervision is the sparse points that experts have already labelled. Three constraints follow from how such a system would be used. No photograph may contribute to both training and testing, because points from one photograph share lighting, camera and reef patch. Performance must be reported at the class frequencies found on the reef, not at artificially balanced ones. And the whole pipeline should train and run on a single free-tier GPU.

3.2 DATA SET

We use the public Coral Reef Dataset on Kaggle (jxwleong/coral-reef-dataset). The image folder holds 4,505 photographs. The annotation file lists 418,310 points on 2,455 of them, and each row gives the photograph name, the row and column of the point, and one of 40 text labels. Two annotation densities are present: most photographs carry 175 to 200 points and a smaller group carries about 20, for a mean of 164.7 points per photograph.

We map the 40 labels to three health classes (Table 3.1). The source has no bleaching labels, so Healthy Coral means live hard or soft coral of any genus. Dead/Degraded groups dead coral and rubble with the algal categories that take over when coral dies. Substrate/CCA is bare or encrusted bottom. Points labelled *off*, *bad* or *dark* mark unusable locations and are dropped, which removes 14,055 rows and leaves 404,255 valid points. Two thirds of them are substrate.

Table 3.1 Mapping of the 40 raw labels to three health classes

Health class	Raw labels	Valid points	Share
Healthy Coral	27 live-coral labels: porites, pocillopora, montipora, pavona, acropora, millepora, soft coral and 20 other genera	68,420	16.9%
Dead / Degraded	dead coral, broken coral rubble, turf, macroalgae, algae, green fleshy algae	68,828	17.0%
Substrate / CCA	crustose coralline algae, hard substrate, rock, sand	267,007	66.0%
Ignored	off, bad, dark	14,055 dropped	

3.3 WHY POINT-CENTRED CROPS

Figure 3.1 shows the label statistics that ruled out whole-image training. Only 78 photographs contain a single class, 706 contain two and 1,671 contain all three. The majority class covers less than half of the points in 9.1% of photographs and at least 90% of them in only 24.8%. Photographs whose majority class is Dead/Degraded are the least pure of all, with a median purity near 0.56.

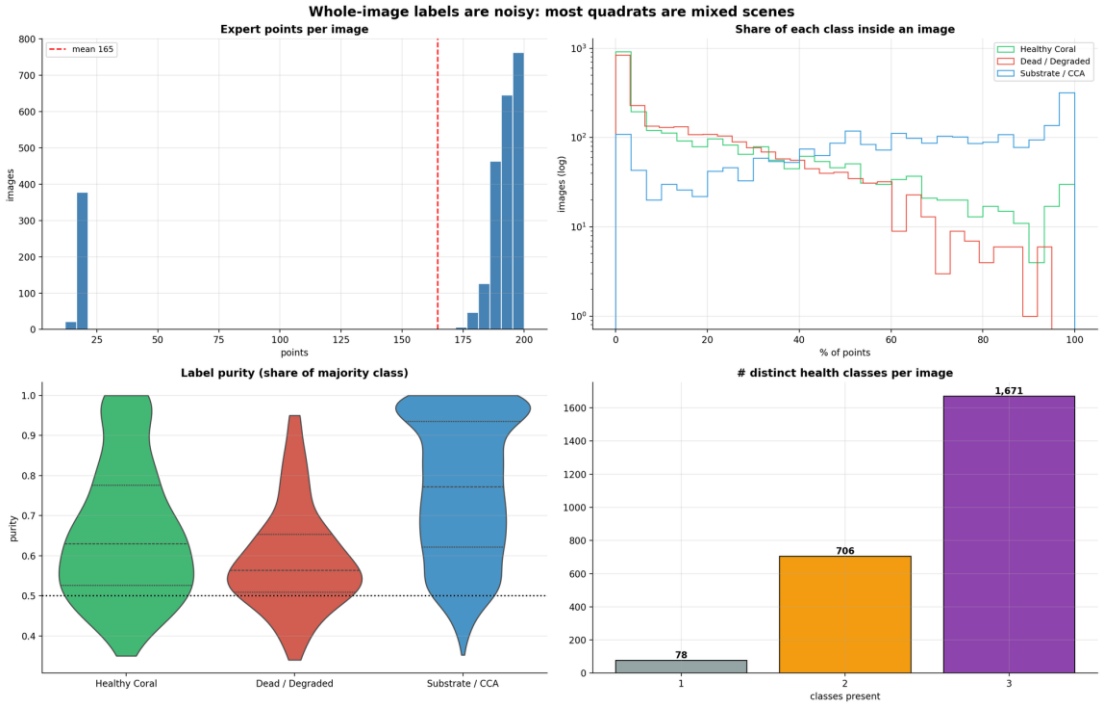


Figure 3.1 Label statistics of the 2,455 annotated quadrats: points per image, class share per image, purity of the majority class, and number of classes present

We therefore make the labelled point the training unit. For each point we cut a 224 × 224 crop centred on it (Figure 3.2). The photograph is first reflect-padded by 112 pixels so that points near the border still get a full crop. The label belongs to the centre pixel, and the rest of the crop is context.

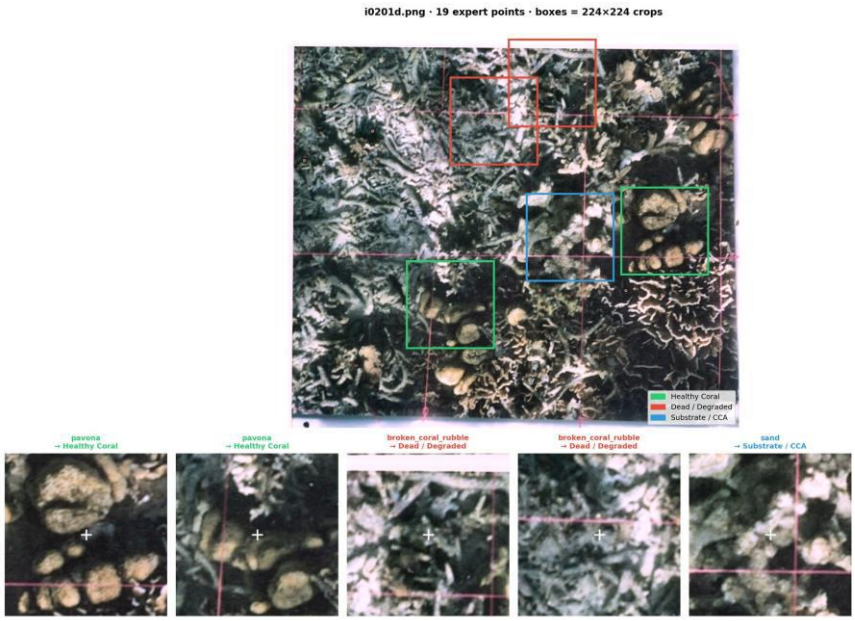


Figure 3.2 A quadrat with its expert points and five of the 224 × 224 crops cut around them; the white cross marks the labelled pixel

3.4 PRE-PROCESSING

Water absorbs red light first, so reef photographs have a blue-green cast that changes with depth and site. Each photograph is corrected once before any crop is cut. A grey-world white balance scales every colour channel so that its mean matches the mean of the three channels, with the gain limited to between 0.5 and 2.5. Contrast-limited adaptive histogram equalisation (clip limit 1.5, 8 × 8 tiles) is then applied to the lightness channel in LAB space. All six models see the same corrected pixels. The crops are stored as 8-bit arrays so that training never decodes a JPEG, and building the cache took 7 minutes.

3.5 DATA SPLIT AND SAMPLING

The photographs are split 70/15/15 into training, validation and test sets, stratified by majority class with a fixed seed (Table 3.2). Splitting by photograph, and not by point, is what prevents leakage: two points 50 pixels apart share most of their crop.

Table 3.2 Grouped split and the crops sampled from each part

Split	Quadrats	Valid points	Crops used	Healthy	Dead	Substrate	Class mix
Train	1,718	285,305	24,000	8,000	8,000	8,000	balanced
Val	368	59,434	5,000	885	913	3,202	natural
Test	369	59,516	6,000	928	1,031	4,041	natural

From the training photographs we draw 8,000 points per class, 24,000 crops in all. The validation and test crops (5,000 and 6,000) are drawn at random from their own photographs and so keep the natural mix of roughly one sixth healthy, one sixth dead and two thirds substrate. Training on balanced data and testing on unbalanced data creates a prior shift, which Section 3.7 deals with.

3.6 EVALUATION PROTOCOLS

3.6.1 Protocol A: Point Crops

Each test crop is classified once. We report accuracy, balanced accuracy, macro-F1, ROC-AUC (one-vs-rest, macro-averaged), Cohen's κ and per-class precision, recall and F1. Accuracy alone is a poor guide here, because a model that always answers Substrate would score about 67%. We treat macro-F1 and the recall of the Dead/Degraded class as the numbers that matter. Calibration is measured by the expected calibration error (Guo et al., 2017) over B = 15 confidence bins,

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|S_b|}{N} |\mathrm{acc}(S_b) - \mathrm{conf}(S_b)| \tag{3.1}$$

where S_b is the set of test points whose confidence falls in bin b and N is the number of test points.

3.6.2 Protocol B: Sliding Window on Whole Quadrats

The classifier is slid over 150 test quadrats to give a dense class map (Section 4.5). The map is scored in two ways: at each of the 25,082 expert points on those quadrats, and as percent cover, where the predicted share of Healthy Coral pixels is compared with the share of expert points labelled Healthy Coral. For cover we report the mean absolute error (MAE) in percentage points, RMSE, Pearson's r and the mean signed error.

3.6.3 Efficiency

We count parameters and FLOPs, and time a single 224 × 224 crop on the GPU in 32-bit precision with the CUDA queue synchronised. Throughput is measured at batch size 64 with mixed precision, and CPU latency on two virtual cores.

3.7 PRIOR-SHIFT CORRECTION

A network trained on equal numbers of each class predicts the two minority classes more often than they occur on the reef. We correct its output probabilities $p_c(x)$ by the ratio of

natural to training priors raised to a power τ, following the logit-adjustment idea of Menon et al. (2021):

$$q_c(x) = \frac{p_c(x)\left(\frac{\pi_c^{\text{nat}}}{\pi_c^{\text{train}}}\right)^{\tau}}{\sum_k p_k(x)\left(\frac{\pi_k^{\text{nat}}}{\pi_k^{\text{train}}}\right)^{\tau}} \tag{3.2}$$

Here $\pi_{\text{train}} = 1/3$ for every class and $\pi_{\text{nat}} = (0.170, 0.168, 0.662)$ is measured on the training photographs only. Setting $\tau = 0$ leaves the prediction unchanged and $\tau = 1$ applies the full Bayes correction. For each model we pick τ from a grid of 0 to 1.5 in steps of 0.1 on the validation set:

$$\tau^* = \arg\max_{\tau} \frac{1}{2}[\text{F1}_{\text{macro}}(\tau) + \text{BA}(\tau)] \tag{3.3}$$

where BA is balanced accuracy. Ties go to the smaller τ. We chose this objective over accuracy because accuracy rewards moving predictions towards the 66% substrate majority, at the expense of the class a reef manager most needs to see. Section 5.4 shows what the choice costs. Test-time augmentation (TTA), the average over four flips of the crop, is evaluated as an option but is not part of the headline numbers.

CHAPTER 4
SYSTEM DESIGN AND EXPERIMENTAL WORK

4.1 PIPELINE OVERVIEW

Figure 4.1 shows the pipeline. One notebook runs every stage in order on a Google Colab instance (one Tesla T4 GPU, two CPU cores, PyTorch 2.11, seed 42): download, label mapping, colour correction, crop caching, the imbalance study, training of the six models, both evaluation protocols, the latency benchmark and the explanation experiments. Every model goes through the same code path, and the only per-model settings are the learning rates, weight decay and number of epochs in Table 4.1.

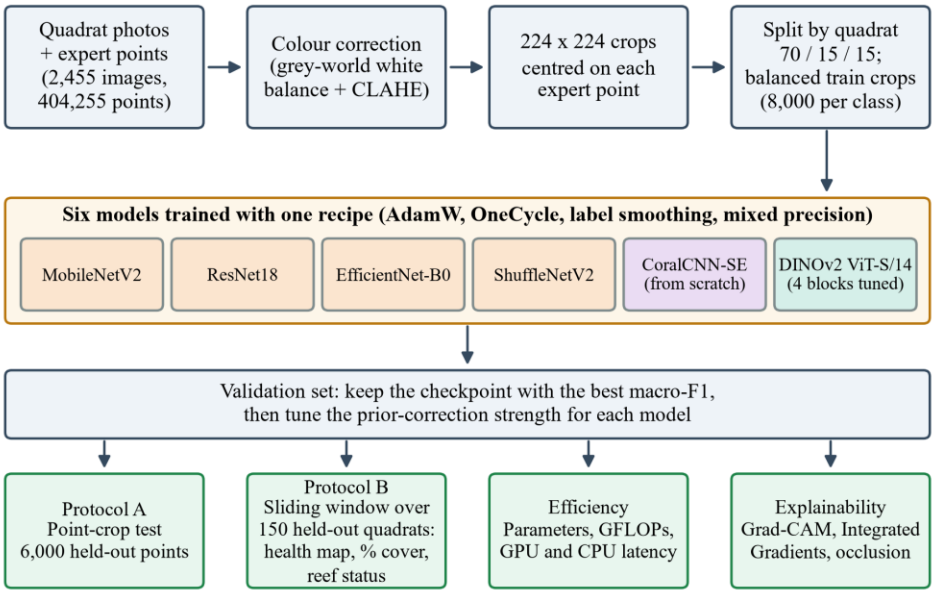


Figure 4.1 Pipeline from annotated quadrats to the four evaluations

4.2 MODELS

Table 4.1 The six models and their per-model training settings

Model	Family	Initial weights	Params (M)	GFLOPs	Max epochs	LR backbone / head	Weight decay
MobileNetV2	Transfer learning	ImageNet	2.23	0.60	12	4.0e-4 / 2.0e-3	0.01
ResNet18	Transfer learning	ImageNet	11.18	3.63	12	3.0e-4 / 1.5e-3	0.01
EfficientNet-B0	Transfer learning	ImageNet	4.01	0.77	12	4.0e-4 / 2.0e-3	0.01
ShuffleNetV2	Transfer learning	ImageNet	1.26	0.29	12	5.0e-4 / 2.0e-3	0.01
CoralCNN-SE	CNN from scratch	random	4.16	2.02	40	2.0e-3 / 2.0e-3	0.02
DINOv2-ViT-S/14	Vision transformer	DINOv2 (LVD-142M)	22.45	11.03	10	1.5e-5 / 1.0e-3	0.05

4.2.1 Transfer-Learning CNNs

MobileNetV2 (Sandler et al., 2018), ResNet18 (He et al., 2016), EfficientNet-B0 (Tan and Le, 2019) and ShuffleNetV2 ×1.0 (Ma et al., 2018) are loaded from torchvision with ImageNet weights. The classifier of each is replaced by dropout (0.2) and a three-way linear layer. All layers are fine-tuned, with the new head at four to five times the learning rate of the backbone.

4.2.2 CoralCNN-SE

CoralCNN-SE is our own network, trained from random initialisation (Figure 4.2). A two-convolution stem is followed by four stages of residual blocks with 64, 128, 192 and 256 channels. The first block of each stage halves the resolution, so a 224-pixel crop ends as a 7 × 7 map. Every block ends in an SE gate with reduction ratio 8. For a block output x with C channels, the gate computes the channel means z, passes them through two 1 × 1 convolutions W_1 and W_2, and scales each channel:

$$s = \sigma\big(W_2\,\delta(W_1\,z)\big), \quad y_c = s_c\,x_c \tag{4.1}$$

where δ is SiLU and σ the sigmoid. The network uses SiLU activations throughout, stochastic depth rising to 0.1 in the last stage, dropout of 0.3 before the classifier and zero-initialised final batch-norm weights in each block. It has 4.16 M parameters, close to EfficientNet-B0.

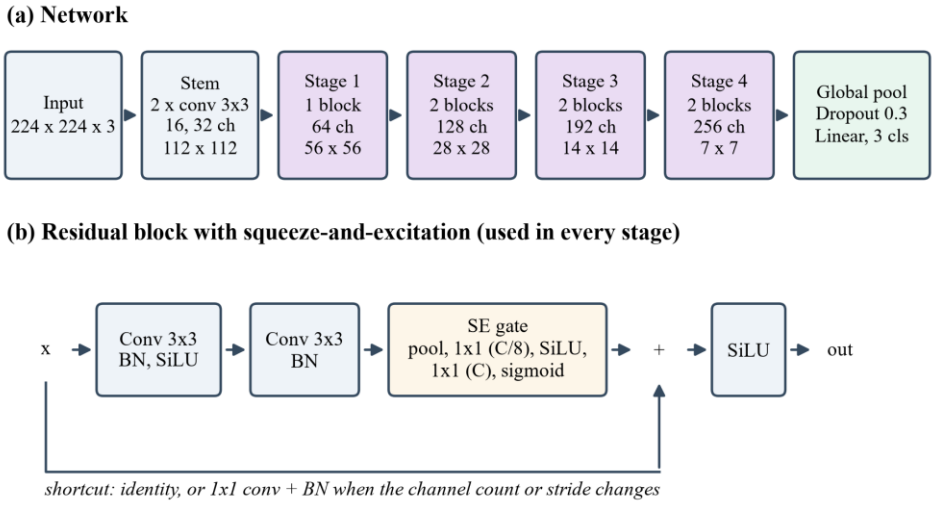


Figure 4.2 CoralCNN-SE: (a) the stages and their output sizes, (b) the residual block with its squeeze-and-excitation gate

4.2.3 DINOv2 ViT-S/14

The transformer is the small DINOv2 model with 14-pixel patches (Oquab et al., 2024), which turns a 224-pixel crop into 256 patch tokens of width 384. We average the patch tokens of each of the last four blocks, concatenate the four averages into a 1,536-dimensional vector, and classify it with a small head (linear 256, layer normalisation, GELU, dropout 0.2, linear 3). At Review 2 the backbone was frozen. In the final pipeline the last four blocks and the final normalisation layer are fine-tuned at a learning rate of 1.5×10^{-5}, which makes about 7.5 M of the 22.45 M parameters trainable.

4.3 TRAINING PROCEDURE

All models are trained with AdamW (Loshchilov and Hutter, 2019) and a one-cycle learning-rate schedule (Smith and Topin, 2018): the rate warms up over the first 20% of steps and then follows a cosine decay. The loss is cross-entropy with label smoothing of 0.1. The batch size is 64, which gives 375 steps per epoch, gradients are clipped at a norm of 5, and mixed precision is on.

After each epoch the model is scored on the validation crops, and the checkpoint with the best validation macro-F1 is kept. Training stops early if macro-F1 has not improved for 5 epochs (transfer CNNs), 12 epochs (CoralCNN-SE) or 4 epochs (DINOv2). Training the six final models took 65.5 minutes in total and the four-recipe study of Section 4.4 about 20 minutes.

Augmentation runs on the GPU, because the Colab instance has only two CPU cores and CPU augmentation was the bottleneck at Review 2. The baseline recipe applies random flips, 90° rotations, a small affine transform (up to 20° rotation, 15% scale and 6% shift, with reflection padding) and mild jitter of brightness, contrast, saturation and per-channel gain. A reef seen from above has no preferred orientation, so flips and rotations do not change the label. Shifts are kept small because the label belongs to the centre pixel.

4.4 CLASS-IMBALANCE RECIPES

Dead/Degraded was the weakest class in every earlier run, so we defined three heavier recipes on top of the baseline R0 (Table 4.2). R1 scales the strength of each transform by class, strongest for Dead/Degraded and lightest for Substrate, and adds wider rotations, shear, gamma changes, a colour cast, blur, sensor noise and random erasing that never covers the centre pixel. R2 adds CutMix (Yun et al., 2019) or MixUp (Zhang et al., 2018a) to half of the

batches. R3 adds a focal term (Lin et al., 2017) and a weight of 1.3 on the Dead/Degraded class.

Table 4.2 Imbalance-handling recipes compared on ResNet18

Recipe	Augmentation	CutMix / MixUp	Loss
R0	Baseline: flips, 90° rotations, mild affine and colour jitter	none	cross-entropy, label smoothing 0.1
R1	Class-aware strong (strength: Healthy 0.8, Dead 1.0, Substrate 0.6)	none	as R0
R2	as R1	applied to 50% of batches	as R0
R3	as R1	applied to 50% of batches	focal ($\gamma = 1$), Dead weight 1.3

Each recipe trains ResNet18 for 8 epochs. The recipe with the best validation score, the objective of Equation 3.3 after tuning τ, is then used for all six models. Test numbers for the recipes are reported but played no part in the choice.

4.5 SLIDING-WINDOW REEF MAPPING

To map a whole quadrat we extract 224 × 224 windows on a regular grid with a stride of 112 pixels (50% overlap) over the reflect-padded photograph, about 281 windows per quadrat. Each window is classified, its probabilities are prior-corrected with the model's τ, and the windows are merged into a dense probability map with Gaussian weights, so that a window counts most near its own centre c_i:

$$P(u) = \frac{\sum_i g(u - c_i)\, q_i}{\sum_i g(u - c_i)}, \quad g(d) = \exp\left(-\frac{\|d\|^2}{2\sigma^2}\right) \tag{4.2}$$

where u is a map position, q_i is the corrected probability vector of window i and σ is 0.35 of the half-window width. The map is computed at a quarter of the image resolution. The class at a position is the arg-max of P(u). Percent cover of a class is the share of map positions assigned to it, and the prediction at an expert point is read from the map at that point.

Live-coral cover is also turned into a four-band reef status of the kind used in monitoring reports: Poor below 10%, Moderate from 10 to 30%, Good from 30 to 50% and Excellent at 50% or more. We count how often the predicted band matches the band given by the expert points.

4.6 EXPLAINABILITY METHODS

Explanations are computed for three models, one per family: ResNet18 (the transfer CNN with the best validation score), CoralCNN-SE and DINOv2. The explained class is always the class the model predicted.

4.6.1 Grad-CAM

Grad-CAM (Selvaraju et al., 2017) weights the feature maps A_k of a chosen layer by the average gradient of the class score y_c and keeps the positive part:

$$\alpha_k^c = \frac{1}{Z}\sum_{i,j}\frac{\partial y^c}{\partial A_{ij}^k}, \quad L^c = \mathrm{ReLU}\left(\sum_k \alpha_k^c A^k\right) \tag{4.3}$$

The layer is the last residual stage for ResNet18 and stage 4 for CoralCNN-SE, both 7 × 7. For DINOv2 we take the output of the last transformer block and reshape its 256 patch tokens to a 16 × 16 grid. Maps are upsampled to 224 × 224.

4.6.2 Integrated Gradients and Occlusion

Integrated Gradients (Sundararajan et al., 2017) attributes the class logit F to each input value by integrating the gradient along a straight path from a reference image x′ to the input x. We use m = 32 midpoint steps and a uniform dark reference image:

$$\mathrm{IG}_i = (x_i - x'_i)\frac{1}{m}\sum_{k=1}^{m}\frac{\partial F\left(x' + \frac{k-0.5}{m}(x - x')\right)}{\partial x_i} \tag{4.4}$$

Occlusion sensitivity (Zeiler and Fergus, 2014) needs no gradients. A 48 × 48 patch of the reference colour is moved over the crop in steps of 24 pixels, and the importance of a pixel is the drop in the class probability averaged over all patches that cover it.

4.6.3 Quantitative Checks

Because each crop is centred on the labelled pixel, a good explanation should put its evidence near the centre. We measure this in two ways. Centre mass is the share of the map's total value that falls within 56 pixels of the centre; a uniform map scores 0.196. The pointing game (Zhang et al., 2018b) asks whether the map's peak lies within 32 pixels of the centre, which a random peak does 6.4% of the time.

Location says nothing about whether the model really uses the highlighted pixels, so we also test faithfulness by perturbation (Petsiuk et al., 2018). In the deletion test the most important pixels are replaced first and the class probability should fall quickly. In the insertion test they are revealed first on a blank image and the probability should rise quickly.

Both curves are run up to 50% of the pixels and summarised by their area. A random ordering is the control. Agreement between two methods is the Spearman rank correlation of their maps after pooling to 14 × 14. The Grad-CAM localisation check uses 300 random test crops and the three-method comparison uses 150.

CHAPTER 5
RESULTS AND DISCUSSION

All numbers in this chapter come from the held-out test photographs. Unless stated otherwise, a model's prediction is a single forward pass followed by the prior correction of Equation 3.2 with the τ tuned on validation data.

5.1 CLASS-IMBALANCE RECIPE STUDY

Table 5.1 Imbalance recipes on ResNet18 (8 epochs each); the recipe is chosen on the validation score

Recipe	τ	Val score	Val Dead recall	Test acc. %	Test macro-F1	Test bal. acc. %	Test Dead recall
R0	0.1	**0.813**	**0.735**	**84.67**	**0.809**	**83.56**	**0.746**
R1	0.0	0.746	0.532	80.58	0.752	76.44	0.583
R2	0.0	0.735	0.398	83.53	0.767	76.29	0.478
R3	0.0	0.758	0.471	83.72	0.776	77.86	0.534

We expected the heavier recipes to help the Dead/Degraded class. They did the opposite (Table 5.1). The baseline R0 has the best validation score, 0.813 against 0.735 to 0.758, and the gap comes almost entirely from Dead/Degraded recall. With CutMix and MixUp (R2) the validation recall of that class falls from 0.735 to 0.398. Focal loss and the class weight in R3 recover part of the loss but stay far below the baseline. The test set tells the same story.

Our reading is as follows. The training crops are already balanced, so re-weighting has no imbalance left to fix. And what separates Dead/Degraded from Substrate is fine texture, a film of turf on rock or the broken edges of rubble. Blur, noise and erasing remove that texture, and mixing two crops makes a label that refers to one centre pixel ambiguous. This is an interpretation, since we did not run the components one at a time, and each recipe had only 8 epochs. R0 was used for all six final models.

5.2 TRAINING BEHAVIOUR

Table 5.2 Training summary of the six final models (validation values are at the saved checkpoint)

Model	Params (M)	Epochs run	Best epoch	Final train acc. %	Val acc. %	Val macro-F1	Train time (min)
MobileNetV2	2.23	12	8	89.3	82.7	0.791	9.3
ResNet18	11.18	12	9	87.8	83.0	0.803	7.0
EfficientNet-B0	4.01	12	7	91.9	**84.7**	0.812	12.8
ShuffleNetV2	1.26	12	9	86.3	80.8	0.780	**6.2**
CoralCNN-SE	4.16	40	34	82.5	77.9	0.760	18.1
DINOv2 ViT-S/14	22.45	10	6	89.8	83.5	**0.816**	12.1

The pretrained models converge within 6 to 9 epochs (Table 5.2, Figure 5.1). DINOv2 has the highest validation macro-F1 from the first epoch and peaks at epoch 6. CoralCNN-SE behaves differently. Its validation curve swings widely for the first 25 epochs, with sharp drops in macro-F1 at epochs 2, 11 and 21 while the learning rate is high, and it settles only as the rate decays. It needs 34 epochs to reach 0.760, 0.02 to 0.06 below the pretrained models on validation, and at 18.1 minutes it is the slowest to train despite having the fastest forward pass.

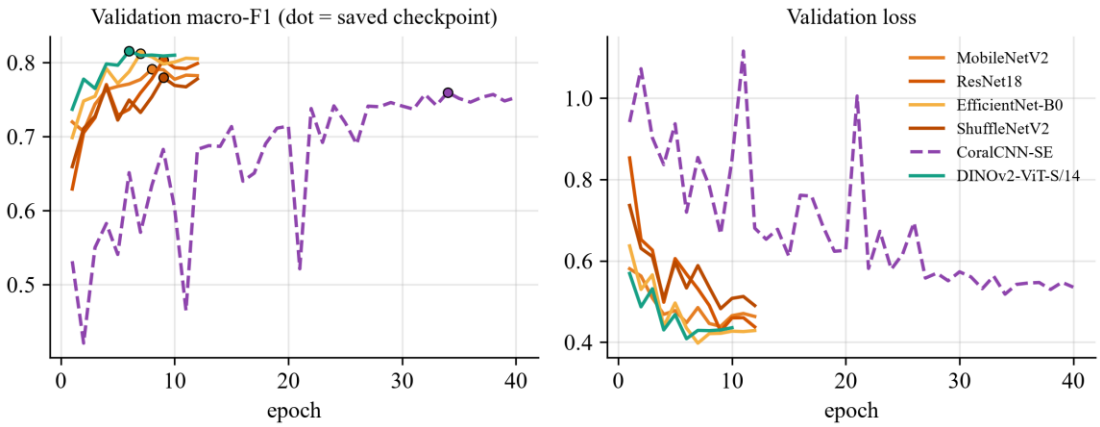


Figure 5.1 Validation macro-F1 and loss per epoch for the six models (redrawn from the training logs)

The gap between training and validation accuracy is roughly 5 to 7 points for every model. Part of it is not overfitting: training accuracy is measured on balanced crops and validation accuracy on the natural mix, before any prior correction.

5.3 POINT-CROP RESULTS

Table 5.3 Protocol A: point-crop metrics on 6,000 held-out expert points (928 Healthy, 1,031 Dead/Degraded, 4,041 Substrate)

							Per-class F1		
Model	**Acc. %**	**Bal. acc. %**	**Macro-F1**	**ROC-AUC**	**κ**	**ECE**	**Healthy**	**Dead**	**Substr.**
MobileNetV2	83.85	81.32	0.792	0.937	0.683	0.033	0.824	0.665	0.887
ResNet18	85.97	82.84	0.819	0.949	0.720	0.038	0.854	0.702	0.902
EfficientNet-B0	85.82	82.40	0.815	0.948	0.715	**0.032**	0.849	0.693	0.902
ShuffleNetV2	83.48	82.29	0.794	0.939	0.682	0.047	0.814	0.685	0.882
CoralCNN-SE	83.68	82.57	0.800	0.939	0.685	0.052	0.843	0.674	0.883
DINOv2 ViT-S/14	**86.70**	**83.94**	**0.831**	**0.956**	**0.734**	0.047	**0.876**	**0.711**	**0.906**

DINOv2 leads on every aggregate metric in Table 5.3, but the margins are small: 0.7 points of accuracy and 0.012 of macro-F1 over ResNet18. All six models fall within 3.2 points of accuracy, from 83.48% to 86.70%. With a single seed we cannot say that differences of a fraction of a point, such as ResNet18 against EfficientNet-B0, would survive a re-run.

The class picture is the same for every model. Substrate is easy (F1 0.88 to 0.91), Healthy Coral is good (0.81 to 0.88), and Dead/Degraded is weak (0.67 to 0.71). The confusion matrices in Figure 5.2 show where the errors go. Between 22% and 29% of Dead/Degraded points are called Substrate, and 6% to 11% of Substrate points are called Dead/Degraded. Confusion between Healthy Coral and Dead/Degraded is rare, 2 to 5% in either direction. The models can tell live coral from everything else. What they find hard is the border between degraded bottom and bare bottom, which is plausible given that turf-covered rock and clean rock differ by a thin film.

CoralCNN-SE deserves a separate note. Trained from scratch, it reaches 83.68% accuracy and a macro-F1 of 0.800. That is ahead of MobileNetV2 and ShuffleNetV2 on macro-F1 and 2.3 points of accuracy behind ResNet18. It has the highest Dead/Degraded recall of all (0.745) and the lowest precision on that class (0.615): it calls 449 substrate points dead, more than any other model. At Review 2 the same design trailed the best pretrained CNN by 18.6 points. The network barely changed between the two runs. The data did, from about 1,700 whole images with majority labels to 24,000 crops with exact labels.



Figure 5.2 Row-normalised confusion matrices on the point-crop test set (counts in brackets)

5.4 PRIOR CORRECTION AND TEST-TIME AUGMENTATION

Table 5.4 Effect of the prior correction (τ tuned on validation) and of four-flip test-time augmentation

		Accuracy %			Macro-			Dead/Degraded recall		
Model	**τ**	**raw**	**+τ**	**+τ +TTA**	**raw**	**+τ**	**+τ +TTA**	**raw**	**+τ**	**+τ +TTA**
MobileNetV2	0.0	83.85	83.85	84.65	0.792	0.792	0.803	0.659	0.659	0.675
ResNet18	0.3	84.40	85.97	86.55	0.808	0.819	0.825	0.761	0.709	0.704
EfficientNet-B0	0.0	85.82	85.82	86.28	0.815	0.815	0.820	0.673	0.673	0.678
ShuffleNetV2	0.2	81.47	83.48	84.95	0.777	0.794	0.812	0.752	0.720	0.743
CoralCNN-SE	0.3	80.90	83.68	84.28	0.780	0.800	0.805	0.816	0.745	0.746
DINOv2 ViT-S/14	0.4	84.50	86.70	87.35	0.817	0.831	0.838	0.800	0.734	0.726

The validation search chose $\tau = 0$ for MobileNetV2 and EfficientNet-B0, which leaves them unchanged, and values of 0.2 to 0.4 for the other four (Table 5.4, Figure 5.3). For those four the correction raises accuracy by 1.6 to 2.8 points and macro-F1 by 0.011 to 0.020, and lowers Dead/Degraded recall by 0.03 to 0.07. This is the expected direction: the correction moves borderline predictions towards the substrate majority.

How far to move them is a decision, not a fact. If τ is tuned for accuracy and macro-F1, the objective we used earlier in the project, DINOv2 picks $\tau = 0.7$ and reaches 87.62% accuracy, but its Dead/Degraded recall falls to 0.660. With the balanced objective it scores

86.70% and 0.734. We kept the balanced objective because missing degraded reef is the costlier mistake in monitoring. A full Bayes correction ($\tau = 1$) is worse than no correction for every model on the validation score.

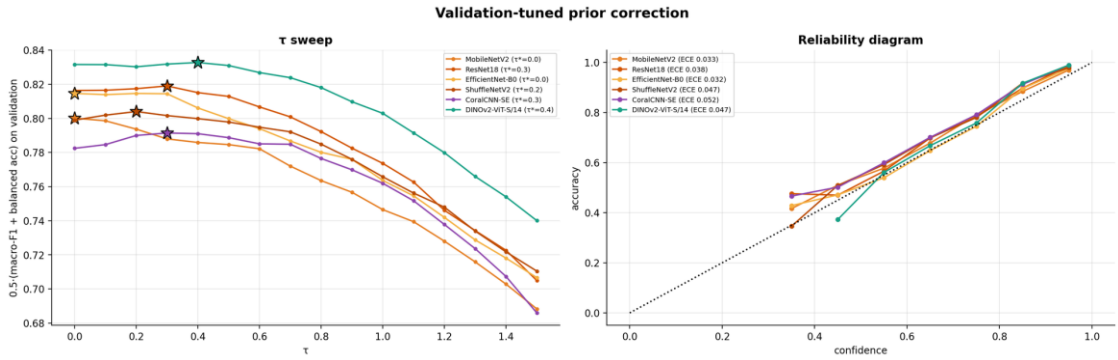


Figure 5.3 Validation score against τ with the chosen value starred (left) and reliability diagram on the test set after correction (right)

Calibration is good after correction, with ECE between 0.032 and 0.052. The reliability curves lie slightly above the diagonal, so the models are a little under-confident, a known effect of label smoothing. Test-time augmentation adds 0.5 to 1.5 points of accuracy for four times the inference cost. With it DINOv2 reaches 87.35% and a macro-F1 of 0.838. We did not use it for the headline numbers or the reef maps.

5.5 EFFICIENCY

Table 5.5 Model size and inference cost for a 224 × 224 crop (Tesla T4; CPU figures on two virtual cores)

Model	Params (M)	Size (MB)	GFLOPs	GPU mean (ms)	GPU p95 (ms)	Throughput (crops/s)	CPU mean (ms)
MobileNetV2	2.23	8.5	0.60	6.12	10.33	2,076	29.3
ResNet18	11.18	42.6	3.63	5.06	6.10	2,459	43.2
EfficientNet-B0	4.01	15.3	0.77	8.84	10.20	1,535	32.1
ShuffleNetV2	**1.26**	**4.8**	**0.29**	7.64	9.12	3,348	**17.5**
CoralCNN-SE	4.16	15.9	2.02	**4.12**	**4.71**	**3,508**	30.4
DINOv2 ViT-S/14	22.45	85.6	11.03	8.50	11.07	678	135.1

No single model is cheapest on every measure (Table 5.5). ShuffleNetV2 is the smallest at 1.26 M parameters and 4.8 MB and the fastest on CPU at 17.5 ms per crop. CoralCNN-SE is the fastest on the GPU, 4.12 ms per crop and about 3,500 crops per second in batches. DINOv2 is the most expensive on every count, and on CPU it needs 135 ms per crop, three to eight times the CNNs.

FLOPs are a poor guide to GPU time, as Ma et al. (2018) argued. EfficientNet-B0 has a fifth of ResNet18's FLOPs and is the slowest CNN on the GPU (8.84 ms against 5.06 ms), because depth-wise convolutions gain little from the GPU. Figure 5.4 plots macro-F1 against latency. On the GPU three models are Pareto-optimal: CoralCNN-SE, ResNet18 and DINOv2. On CPU the front also includes ShuffleNetV2 and EfficientNet-B0. MobileNetV2 is dominated in both.

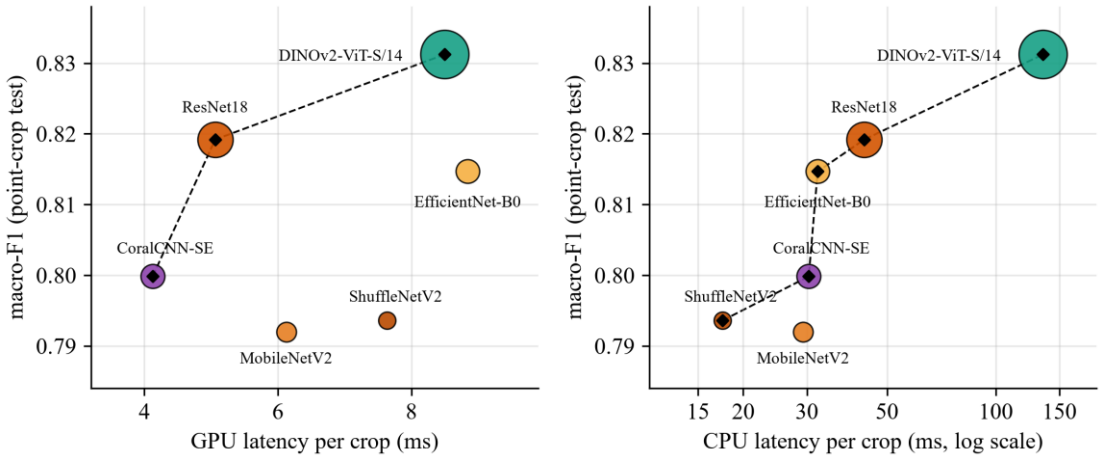


Figure 5.4 Macro-F1 against GPU and CPU latency per crop; marker area follows parameter count, diamonds mark Pareto-optimal models (redrawn from Tables 5.3 and 5.5)

5.6 SLIDING-WINDOW REEF MAPS

Table 5.6 Protocol B: sliding window (stride 112) on 150 held-out quadrats, scored at 25,082 expert points and as percent cover

	At expert points						Live-coral cover			
Model	**Acc. %**	**Macro-F1**	**Dead recall**	**MAE (pp)**	**RMSE (pp)**	**r**	**Bias (pp)**	**Dead cover MAE (pp)**	**Status agree %**	**ms per quadrat**
MobileNetV2	82.49	0.773	0.600	5.54	9.68	0.939	+3.24	5.91	80.7	174
ResNet18	84.64	0.803	0.655	3.23	6.34	0.972	-1.15	5.12	**92.7**	140
EfficientNet-B0	84.10	0.795	0.631	3.72	7.40	0.959	+0.39	5.61	89.3	235
ShuffleNetV2	83.75	0.795	0.667	3.66	6.06	0.973	+0.76	4.57	90.7	115
CoralCNN-SE	82.86	0.792	**0.714**	3.51	6.94	0.964	-0.09	5.41	90.7	**99**
DINOv2 ViT-S/14	**84.82**	**0.805**	0.647	**3.11**	**5.17**	**0.981**	+0.15	**4.52**	92.0	506

Table 5.6 gives the whole-quadrat results and Figure 5.5 shows four of the maps. At the expert points five of the six models lose 0.8 to 1.9 points of accuracy compared with Protocol A, and only ShuffleNetV2 gains (0.3 points). That is expected. The windows sit on a fixed grid and are not centred on the points, and blending smooths away small patches. The top

three keep their order, with DINOv2 at 84.82% and ResNet18 at 84.64%. Dead/Degraded recall drops more, to between 0.60 and 0.71, since dead patches are often small.

For monitoring, the cover estimate matters more than the per-point score. Five of the six models estimate live-coral cover to within 3.1 to 3.7 percentage points on average, with correlations of 0.96 to 0.98 against the expert count and a bias of about one point or less (Figure 5.6). MobileNetV2 is the exception: it over-predicts coral by 3.2 points on average. DINOv2 is the most accurate at 3.11 points, and ResNet18 is close at 3.23.

An error of three points should be read against the reference itself. The expert figure is a count of about 165 to 200 points, and for a quadrat with 20% coral the sampling error of such a count is close to three points. The maps are therefore about as close to the expert cover as a second point count would be. Dead/Degraded cover is estimated less well, with an MAE of 4.5 to 5.9 points, and five of the six models under-estimate it.

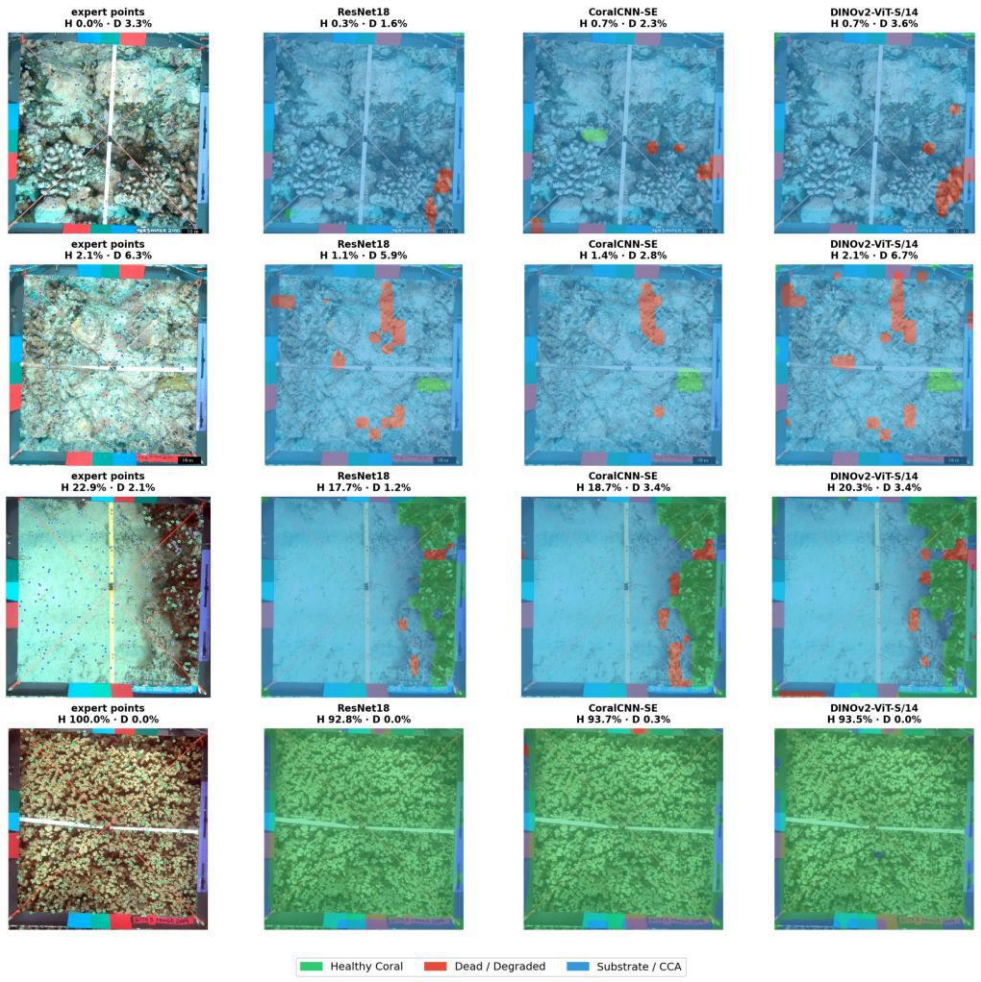


Figure 5.5 Dense health maps for four held-out quadrats spanning 0% to 100% expert coral cover; expert points on the left, three of the six models shown, with predicted Healthy (H) and Dead (D) cover above each map

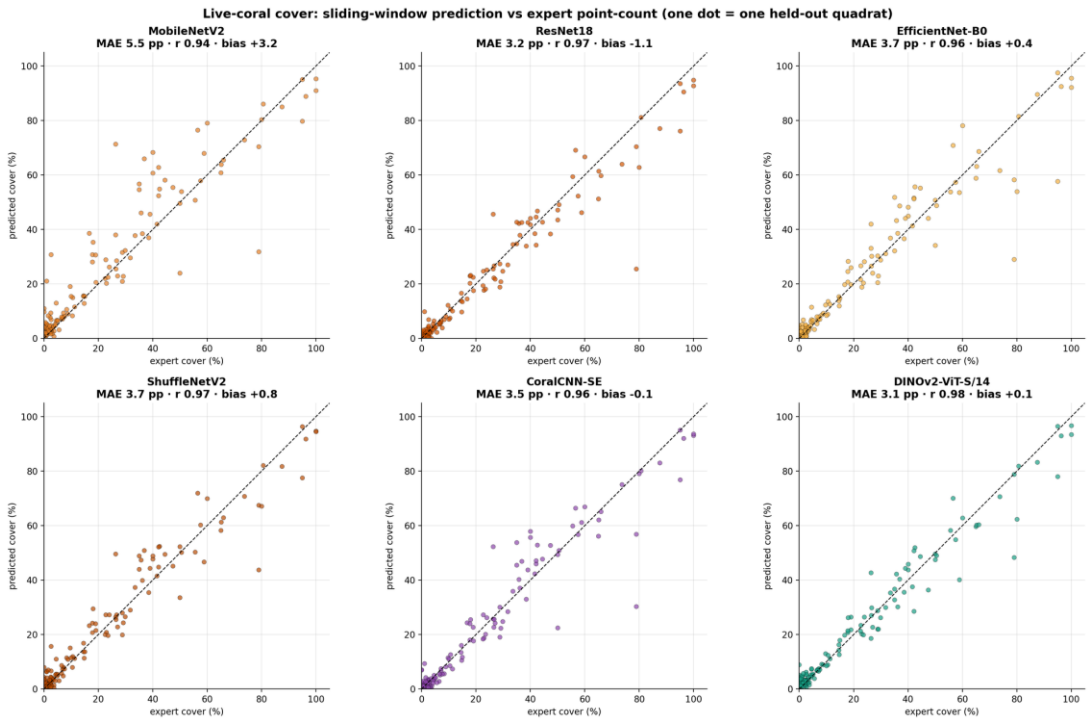


Figure 5.6 Predicted against expert live-coral cover for the 150 held-out quadrats (one dot per quadrat)

The predicted reef-status band matches the expert band for 92.7% of quadrats with ResNet18 and 92.0% with DINOv2, and for 80.7% with MobileNetV2. Mapping a quadrat takes 99 ms with CoralCNN-SE, 140 ms with ResNet18 and 506 ms with DINOv2. One caveat applies to all cover figures: the map includes the quadrat frame and tape visible in Figure 5.5, which the models mostly call Substrate. We did not mask them out.

5.7 EFFECT OF WINDOW STRIDE

Table 5.7 Stride study with DINOv2 on a 20-quadrat subset

Stride (px)	Overlap %	Windows per quadrat	Point acc. %	Macro-F1	Cover MAE (pp)	ms per quadrat
224	0	75	83.19	0.791	3.52	178
112	50	292	87.67	0.849	2.78	530
56	75	1138	89.58	0.873	2.93	1,835

Table 5.7 varies the stride for DINOv2 on 20 quadrats. Because this is a subset, its numbers are not comparable with Table 5.6. More overlap steadily improves the per-point score, from 83.2% with no overlap to 89.6% at a stride of 56 pixels. Cover error does not follow: it is 2.78 points at stride 112 and 2.93 at stride 56, for 3.5 times the computation. A stride of 112 is enough for cover estimation. A finer stride is worth its cost only when the map itself is the product.

5.8 EXPLAINABILITY

Table 5.8 Grad-CAM localisation and deletion test on 300 test crops (a uniform map has centre mass 0.196; a random peak hits 6.4%)

Model	Explained layer	Centre mass	Pointing hit %	Drop, top 20% erased	Drop, random 20%	Ratio
ResNet18	last stage (7×7)	**0.550**	**91.3**	**0.240**	0.038	**6.24**
CoralCNN-SE	stage4 (7×7)	0.272	22.7	0.169	0.082	2.06
DINOv2 ViT-S/14	last block tokens (16×16)	0.218	10.7	0.129	0.022	5.84

Figure 5.7 shows Grad-CAM maps for the same six test crops through the three models, and Table 5.8 puts numbers on what the picture suggests. ResNet18 produces a compact blob on the centre of the crop. Its peak lies within 32 pixels of the labelled point in 91.3% of crops, against 6.4% for a random peak, and erasing the top 20% of its map lowers the class probability six times more than erasing a random 20%. CoralCNN-SE spreads its evidence over textured regions and hits the point in 22.7% of crops. For DINOv2 the Grad-CAM maps are patchy, their centre mass (0.218) is barely above that of a uniform map, and the peak hits the point in 10.7% of crops.

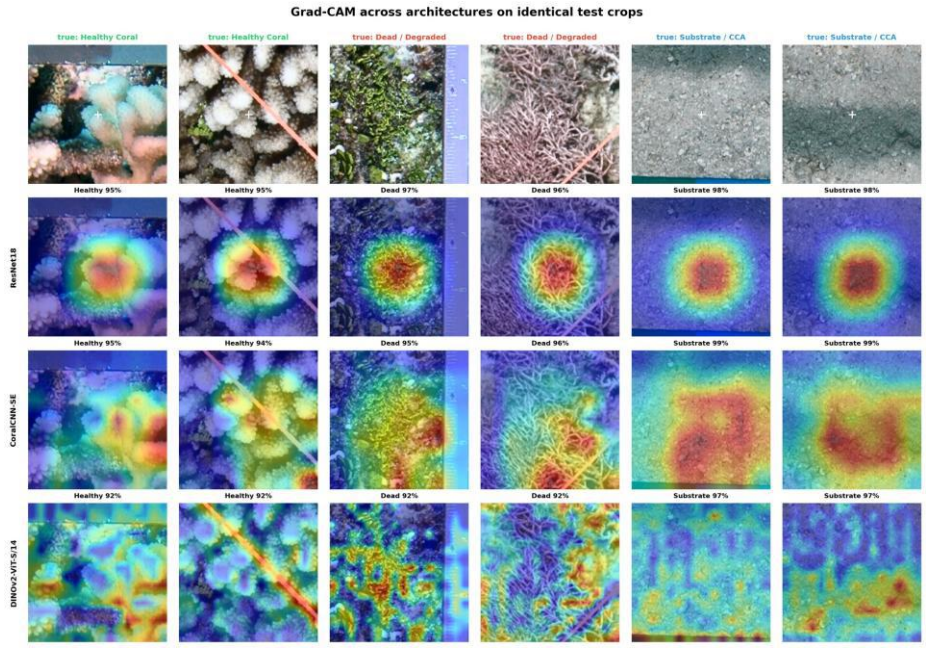


Figure 5.7 Grad-CAM for the same six test crops through ResNet18, CoralCNN-SE and DINOv2; the white cross in the top row is the labelled point

Two cautions apply. A centred map shows that ResNet18 attends to the right place. It does not show that the features it reads there are the ecologically right ones, and a model that

had only learned “the answer is in the middle” would score well on the pointing game too. And the poor DINOv2 result is a statement about Grad-CAM on this model, not about the model, whose classifier averages tokens from four blocks. Panboonyuen (2026) found the same degradation for a ViT on CT scans.

Table 5.9 and Figure 5.8 compare Grad-CAM with Integrated Gradients and occlusion on 150 crops. Occlusion has the largest insertion-minus-deletion score for all three models. Grad-CAM is clearly better than random for ResNet18 (+0.065 against −0.220), marginal for CoralCNN-SE, and close to random for DINOv2 (−0.185 against −0.258). Integrated Gradients is negative for all three. Its completeness error, the mismatch between the summed attributions and the change in the logit, is 0.13 for ResNet18, 0.42 for CoralCNN-SE and 2.28 for DINOv2, so with 32 steps the integral has not converged for the transformer and its IG maps should not be trusted.

Table 5.9 Three explanation methods compared on 150 test crops; a faithful method has low deletion AUC, high insertion AUC and a large difference

Model	Method	Deletion AUC	Insertion AUC	Insertion − Deletion	Centre mass	Pointing hit %
ResNet18	Grad-CAM	0.653	0.719	+0.065	0.556	93.3
	Integrated Gradients	0.605	0.589	-0.016	0.418	58.7
	Occlusion	0.546	0.755	**+0.208**	0.433	40.7
	Random	0.747	0.527	-0.220	0.191	3.3
CoralCNN-SE	Grad-CAM	0.581	0.596	+0.015	0.282	24.7
	Integrated Gradients	0.552	0.481	-0.071	0.354	28.0
	Occlusion	0.521	0.598	**+0.076**	0.403	34.0
	Random	0.642	0.433	-0.209	0.198	3.3
DINOv2 ViT-S/14	Grad-CAM	0.668	0.483	-0.185	0.208	10.0
	Integrated Gradients	0.704	0.507	-0.197	0.274	38.7
	Occlusion	0.595	0.674	**+0.079**	0.343	30.0
	Random	0.787	0.529	-0.258	0.198	6.0

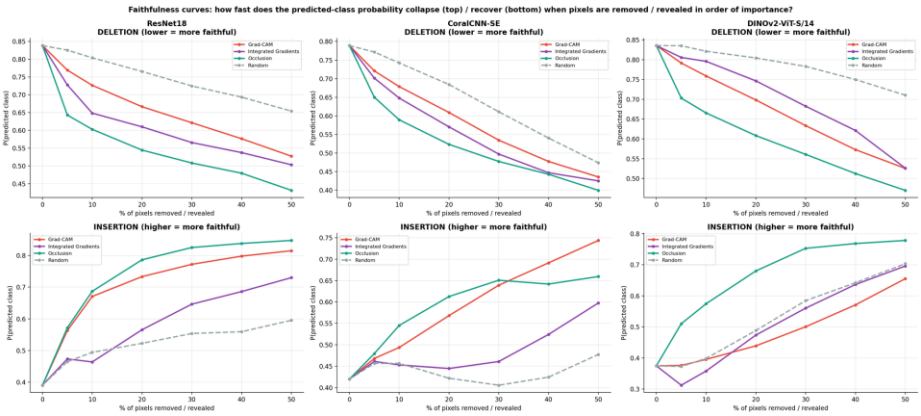


Figure 5.8 Deletion (top) and insertion (bottom) curves for the three explanation methods and a random ordering

The methods also disagree with each other more as the architecture moves away from a plain CNN. The rank correlation between Grad-CAM and Integrated Gradients is 0.67 for ResNet18, 0.32 for CoralCNN-SE and 0.07 for DINOv2. The pointing accuracy for ResNet18 is 93.3% in Table 5.9 and 91.3% in Table 5.8 only because the two experiments drew different crops.

5.9 OVERALL COMPARISON

Table 5.10 Overall comparison across both protocols and cost

	Accuracy				Cost				Rank		
Model	**Crop acc. %**	**Crop F1**	**Map F1**	**Cover MAE**	**Params (M)**	**GPU ms**	**CPU ms**	**ms per quadrat**	**Acc. rank**	**Eff. rank**	**Overall**
MobileNetV2	83.85	0.792	0.773	5.54	2.23	6.12	29.3	174	5.50	2.33	6
ResNet18	85.97	0.819	0.803	3.23	11.18	5.06	43.2	140	2.00	4.00	2
EfficientNet-B0	85.82	0.815	0.795	3.72	4.01	8.84	32.1	235	3.75	4.33	4
ShuffleNetV2	83.48	0.794	0.795	3.66	**1.26**	7.64	**17.5**	115	4.50	**2.00**	5
CoralCNN-SE	83.68	0.800	0.792	3.51	4.16	**4.12**	30.4	**99**	4.25	2.67	3
DINOv2 ViT-S/14	**86.70**	**0.831**	**0.805**	**3.11**	22.45	8.50	135.1	506	**1.00**	5.67	**1**

Table 5.10 brings the criteria together. The accuracy rank is the mean rank over crop accuracy, crop macro-F1, map macro-F1 and cover MAE. The efficiency rank is the mean rank over GPU latency, CPU latency and parameter count. The overall rank weights accuracy four to one against GPU latency. On that basis the order is DINOv2, ResNet18, CoralCNN-SE, EfficientNet-B0, ShuffleNetV2 and MobileNetV2.

The ranking should not hide how close the top two are where it counts. ResNet18 estimates cover to 3.23 points against 3.11 for DINOv2, agrees with the expert status band slightly more often, and maps a quadrat 3.6 times faster with half the parameters. For a

survey team processing thousands of quadrats on modest hardware it is the more sensible choice. DINOv2 is the one to pick when the last point of accuracy matters and a GPU is available, and ShuffleNetV2 or CoralCNN-SE when the model must run on a CPU.

Table 5.11 Review 2 results against the final pipeline (final test unit: 6,000 point crops)

		Review 2		Final		
Model	**Review 2 test unit**	**Acc. %**	**Macro-F1**	**Acc. %**	**Macro-F1**	**Change in acc.**
MobileNetV2	whole image (369)	81.0	0.709	83.85	0.792	+2.9
ResNet18	whole image (369)	82.4	0.732	85.97	0.819	+3.6
EfficientNet-B0	whole image (369)	82.1	0.704	85.82	0.815	+3.7
ShuffleNetV2	whole image (369)	84.3	0.763	83.48	0.794	-0.8
CoralCNN-SE	whole image (361)	65.7	0.524	83.68	0.800	+18.0
DINOv2 ViT-S/14	point crop (4000)	77.7	not reported	86.70	0.831	+9.0

Table 5.11 sets the final results beside those of Review 2. The comparison is indicative and not strict, because the earlier CNNs were tested on whole images with majority labels. With that caveat, five of the six models gained accuracy, and the two that gained most are the ones that were most limited by the old setup: CoralCNN-SE by 18 points and DINOv2 by 9, the latter after its last four blocks were unfrozen. Macro-F1 rose for all CNNs, from 0.52 to 0.76 at Review 2 to 0.79 to 0.82 now. ShuffleNetV2, the best model at Review 2, is the one that lost accuracy (0.8 points) and is now last on crop accuracy, although its macro-F1 rose from 0.763 to 0.794.

5.10 LIMITATIONS

Several limits qualify these results. Dead/Degraded is still the weak class, with F1 no higher than 0.71, and part of the cause is our own mapping, which puts dead coral, rubble, turf and macroalgae under one name. All photographs come from one public collection with no independent site or season, a setting in which Shao et al. (2025) and Felemban et al. (2025) report large drops. Each model was trained once with one seed, so differences smaller than about a point between models are not established.

CoralCNN-SE was not trained without its SE gates, so the contribution of the gate is unknown. Cover is computed over the full photograph, quadrat frame included, against an expert figure that is itself a sample of 20 to 200 points.

CHAPTER 6
CONCLUSIONS AND SCOPE FOR FURTHER WORK

6.1 CONCLUSIONS

The project set out to predict reef health from quadrat photographs that carry only sparse expert points. Moving from whole-image labels to crops centred on those points was the change that mattered most. It gave 24,000 clean training examples in place of 1,700 noisy ones and raised the macro-F1 of every CNN. The largest gain went to the CNN trained from scratch, which rose from 65.7% to 83.7% accuracy and now sits within 2.3 points of ResNet18.

On 6,000 held-out points the fine-tuned DINOv2 ViT-S/14 is the most accurate model at 86.70% accuracy and a macro-F1 of 0.831, with ResNet18 at 85.97% and 0.819. Slid over whole quadrats, the same classifiers estimate live-coral cover to within about three percentage points of the expert count, roughly the sampling error of the count itself, and agree with the expert reef-status band for 92 to 93% of quadrats. ResNet18 does this in 140 ms per quadrat and DINOv2 in 506 ms.

Three findings went against our expectations. Heavier augmentation, CutMix/MixUp and focal loss reduced the recall of the Dead/Degraded class instead of raising it. A prior correction helped accuracy but had to be tuned for a balanced objective, or it traded away that same class. And Grad-CAM, which localises the labelled point well for ResNet18, is close to uninformative for the transformer, where only occlusion sensitivity passed the faithfulness tests. The weak point has not changed since the start of the project: every model confuses Dead/Degraded bottom with bare substrate, and nothing we tried moved its F1 above 0.71.

6.2 SCOPE FOR FURTHER WORK

1. Split Dead/Degraded into dead coral, rubble and algae, or predict the label hierarchy as Blondin et al. (2024) do, and merge to health classes at the end.

2. Test on a second reef collection, first without retraining and then with a little adaptation, to measure the cross-site drop.

3. Train CoralCNN-SE with and without SE gates, and repeat all runs over several seeds to put error bars on the comparison.

4. Mask the quadrat frame before computing cover, and compare the cover error with the disagreement between two human annotators.

5. Compare partial fine-tuning of DINOv2 with the low-rank adapters of Shao et al. (2025), and try explanation methods built for transformers.

6. Package ResNet18 and DINOv2 behind a small inference service that returns the map, the cover figures and the reef-status band for an uploaded quadrat.

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APPENDIX

Appendix A: Plagiarism Report

Insert the plagiarism report of this document here before submission.

Appendix B: Reproducibility

The complete pipeline is in the notebook *coral_reef_final_pipeline_v2.ipynb*. The results archive *coral_reef_final_results.zip* holds the weights of the six models, the per-epoch logs, every table of Chapter 5 as a CSV file, the class and probabilities of all 252,756 sliding windows, and the per-quadrat composition of the 150 test quadrats for every model. The file *results_master.json* in the archive lists all metrics quoted in this report.